Test ID	Category	DType	i_data	i_scale	i_zero_point	Expected Output	Overflow	Description	Result
TC1.1	Reset	All inputs = 0	0	hex 0x00000000	0	0x00 (0)	0	Hold reset low for two rising edges.	Pass
TC2.1	int8 specific test cases	int8 (0)	23'sd162574	hex 0x391C68A3	8'd41	0x41 (65)	0	normal positive, expected 65	Pass
TC2.2	int8 specific test cases	int8 (0)	23'sd147	hex 0x3E7182C0	-8'sd12	0x17 (23)	0	negative zero point, expected 23	Pass
TC2.3	int8 specific test cases	int8 (0)	-23'sd9	hex 0x415AAAAB	8'd108	0xF1 (-15)	0	negative data, expected -15 = 8'hF1	Pass
TC2.4	int8 specific test cases	int8 (0)	23'sd0	hex 0x45841234	8'd124	0x7C (124)	0	zero data, expected 124	Pass
TC2.5	int8 specific test cases	int8 (0)	23'sd1	hex 0x45841234	8'd124	0x7F (127)	1	large positive overflow, expected 127, overflow	Pass
TC2.6	int8 specific test cases	int8 (0)	23'sd1	hex 0x43800000	8'd124	0x7F (127)	1	positive overflow, expected 127, overflow	Pass
TC2.7	int8 specific test cases	int8 (0)	-23'sd1	hex 0x43800000	8'd124	0x80 (-128)	1	exponent > 134 causes negative early saturation; zero point is bypassed, expected -128 = 8'h80, overflow	Pass
TC2.8	int8 specific test cases	int8 (0)	23'sd127	hex 0x3F800000	8'd0	0x7F (127)	0	exact positive edge, expected 127	Pass
TC2.9	int8 specific test cases	int8 (0)	23'sd128	hex 0x3F800000	8'd0	0x7F (127)	1	one above positive edge, expected 127, overflow	Pass
TC2.10	int8 specific test cases	int8 (0)	-23'sd128	hex 0x3F800000	8'd0	0x80 (-128)	0	exact negative edge, expected -128 = 8'h80	Pass
TC2.11	int8 specific test cases	int8 (0)	-23'sd129	hex 0x3F800000	8'd0	0x80 (-128)	1	one below negative edge, expected -128, overflow	Pass
TC2.12	int8 specific test cases	int8 (0)	23'sd8	hex 0x3D800000	8'd0	0x00 (0)	0	+0.5 tie, expected 0	Pass
TC2.13	int8 specific test cases	int8 (0)	23'sd4	hex 0x3EC00000	8'd0	0x02 (2)	0	+1.5 tie, expected 2	Pass
TC2.14	int8 specific test cases	int8 (0)	-23'sd8	hex 0x3D800000	8'd0	0x00 (0)	0	-0.5 tie, expected 0	Pass
TC2.15	int8 specific test cases	int8 (0)	-23'sd4	hex 0x3EC00000	8'd0	0xFE (-2)	0	-1.5 tie, expected -2 = 8'hFE	Pass
TC3.1	uint8 specific test cases	uint8 (1)	23'sd0	hex 0x3F800000	8'd0	0x00 (0)	0	exact low edge, expected 0	Pass
TC3.2	uint8 specific test cases	uint8 (1)	23'sd255	hex 0x3F800000	8'd0	0xFF (255)	0	exact high edge, expected 255	Pass
TC3.3	uint8 specific test cases	uint8 (1)	23'sd256	hex 0x3F800000	8'd0	0xFF (255)	1	one above high edge, expected 255, overflow	Pass
TC3.4	uint8 specific test cases	uint8 (1)	-23'sd1	hex 0x3F800000	8'd0	0x00 (0)	1	below low edge, expected 0, overflow	Pass
TC3.5	uint8 specific test cases	uint8 (1)	23'sd250	hex 0x3F800000	8'd5	0xFF (255)	0	250 + 5 = 255, expected 255	Pass
TC3.6	uint8 specific test cases	uint8 (1)	23'sd251	hex 0x3F800000	8'd5	0xFF (255)	1	251 + 5 = 256, expected 255, overflow	Pass
TC3.7	uint8 specific test cases	uint8 (1)	-23'sd5	hex 0x3F800000	8'd5	0x00 (0)	0	-5 + 5 = 0, expected 0	Pass
TC3.8	uint8 specific test cases	uint8 (1)	-23'sd6	hex 0x3F800000	8'd5	0x00 (0)	1	-6 + 5 = -1, expected 0, overflow	Pass
TC3.9	uint8 specific test cases	uint8 (1)	23'sd1	hex 0x3F000000	8'd0	0x00 (0)	0	0.5 tie, expected 0	Pass
TC3.10	uint8 specific test cases	uint8 (1)	23'sd3	hex 0x3F000000	8'd0	0x02 (2)	0	1.5 tie, expected 2	Pass
TC3.11	uint8 specific test cases	uint8 (1)	-23'sd1	hex 0x3F000000	8'd0	0x00 (0)	0	-0.5 tie, expected 0	Pass
TC3.12	uint8 specific test cases	uint8 (1)	-23'sd3	hex 0x3F000000	8'd0	0x00 (0)	1	-1.5 tie, expected 0, overflow	Pass
TC4.1	int4 specific test cases	int4 (2)	-23'sd8	hex 0x3F800000	8'd0	0xF8 (-8)	0	exact negative edge, expected -8 = 8'hF8	Pass
TC4.2	int4 specific test cases	int4 (2)	-23'sd9	hex 0x3F800000	8'd0	0xF8 (-8)	1	one below negative edge, expected -8, overflow	Pass
TC4.3	int4 specific test cases	int4 (2)	23'sd7	hex 0x3F800000	8'd0	0x07 (7)	0	exact positive edge, expected 7	Pass
TC4.4	int4 specific test cases	int4 (2)	23'sd8	hex 0x3F800000	8'd0	0x07 (7)	1	one above positive edge, expected 7, overflow	Pass
TC4.5	int4 specific test cases	int4 (2)	23'sd3	hex 0x3F800000	8'd4	0x07 (7)	0	3 + 4 = 7, expected 7	Pass
TC4.6	int4 specific test cases	int4 (2)	23'sd4	hex 0x3F800000	8'd4	0x07 (7)	1	4 + 4 = 8, expected 7, overflow	Pass
TC4.7	int4 specific test cases	int4 (2)	-23'sd3	hex 0x3F800000	-8'sd5	0xF8 (-8)	0	-3 - 5 = -8, expected -8	Pass
TC4.8	int4 specific test cases	int4 (2)	-23'sd4	hex 0x3F800000	-8'sd5	0xF8 (-8)	1	-4 - 5 = -9, expected -8, overflow	Pass
TC4.9	int4 specific test cases	int4 (2)	23'sd1	hex 0x3F000000	8'd0	0x00 (0)	0	0.5 tie, expected 0	Pass
TC4.10	int4 specific test cases	int4 (2)	23'sd3	hex 0x3F000000	8'd0	0x02 (2)	0	1.5 tie, expected 2	Pass
TC4.11	int4 specific test cases	int4 (2)	-23'sd1	hex 0x3F000000	8'd0	0x00 (0)	0	-0.5 tie, expected 0	Pass
TC4.12	int4 specific test cases	int4 (2)	-23'sd3	hex 0x3F000000	8'd0	0xFE (-2)	0	-1.5 tie, expected -2 = 8'hFE	Pass
TC5.1	uint4 specific test cases	uint4 (3)	23'sd0	hex 0x3F800000	8'd0	0x00 (0)	0	exact low edge, expected 0	Pass

TC5.2	uint4 specific test cases	uint4 (3)	23'sd15	hex 0x3F800000	8'd0	0x0F (15)	0	exact high edge, expected 15 = 8'h0F	Pass
TC5.3	uint4 specific test cases	uint4 (3)	23'sd16	hex 0x3F800000	8'd0	0x0F (15)	1	one above high edge, expected 15, overflow	Pass
TC5.4	uint4 specific test cases	uint4 (3)	-23'sd1	hex 0x3F800000	8'd0	0x00 (0)	1	below low edge, expected 0, overflow	Pass
TC5.5	uint4 specific test cases	uint4 (3)	23'sd10	hex 0x3F800000	8'd5	0x0F (15)	0	10 + 5 = 15, expected 15	Pass
TC5.6	uint4 specific test cases	uint4 (3)	23'sd11	hex 0x3F800000	8'd5	0x0F (15)	1	11 + 5 = 16, expected 15, overflow	Pass
TC5.7	uint4 specific test cases	uint4 (3)	-23'sd5	hex 0x3F800000	8'd5	0x00 (0)	0	-5 + 5 = 0, expected 0	Pass
TC5.8	uint4 specific test cases	uint4 (3)	-23'sd6	hex 0x3F800000	8'd5	0x00 (0)	1	-6 + 5 = -1, expected 0, overflow	Pass
TC5.9	uint4 specific test cases	uint4 (3)	23'sd1	hex 0x3F000000	8'd0	0x00 (0)	0	0.5 tie, expected 0	Pass
TC5.10	uint4 specific test cases	uint4 (3)	23'sd3	hex 0x3F000000	8'd0	0x02 (2)	0	1.5 tie, expected 2	Pass
TC5.11	uint4 specific test cases	uint4 (3)	23'sd31	hex 0x3F000000	8'd0	0x0F (15)	1	15.5 tie -> 16, expected 15, overflow	Pass
TC5.12	uint4 specific test cases	uint4 (3)	-23'sd1	hex 0x3F000000	8'd0	0x00 (0)	0	-0.5 tie, expected 0	Pass
TC5.13	uint4 specific test cases	uint4 (3)	-23'sd3	hex 0x3F000000	8'd0	0x00 (0)	1	-1.5 tie, expected 0, overflow	Pass
TC6.1	int2 specific test cases	int2 (4)	-23'sd2	hex 0x3F800000	8'd0	0xFE (-2)	0	exact negative edge, expected -2 = 8'hFE	Pass
TC6.2	int2 specific test cases	int2 (4)	-23'sd3	hex 0x3F800000	8'd0	0xFE (-2)	1	one below negative edge, expected -2, overflow	Pass
TC6.3	int2 specific test cases	int2 (4)	23'sd1	hex 0x3F800000	8'd0	0x01 (1)	0	exact positive edge, expected 1	Pass
TC6.4	int2 specific test cases	int2 (4)	23'sd2	hex 0x3F800000	8'd0	0x01 (1)	1	one above positive edge, expected 1, overflow	Pass
TC6.5	int2 specific test cases	int2 (4)	23'sd2	hex 0x3F800000	-8'sd1	0x01 (1)	0	2 - 1 = 1, expected 1	Pass
TC6.6	int2 specific test cases	int2 (4)	23'sd3	hex 0x3F800000	-8'sd1	0x01 (1)	1	3 - 1 = 2, expected 1, overflow	Pass
TC6.7	int2 specific test cases	int2 (4)	-23'sd1	hex 0x3F800000	-8'sd1	0xFE (-2)	0	-1 - 1 = -2, expected -2	Pass
TC6.8	int2 specific test cases	int2 (4)	-23'sd2	hex 0x3F800000	-8'sd1	0xFE (-2)	1	-2 - 1 = -3, expected -2, overflow	Pass
TC6.9	int2 specific test cases	int2 (4)	23'sd1	hex 0x3F000000	8'd0	0x00 (0)	0	0.5 tie, expected 0	Pass
TC6.10	int2 specific test cases	int2 (4)	23'sd3	hex 0x3F000000	8'd0	0x01 (1)	1	1.5 tie -> 2, expected 1, overflow	Pass
TC6.11	int2 specific test cases	int2 (4)	-23'sd1	hex 0x3F000000	8'd0	0x00 (0)	0	-0.5 tie, expected 0	Pass
TC6.12	int2 specific test cases	int2 (4)	-23'sd3	hex 0x3F000000	8'd0	0xFE (-2)	0	-1.5 tie, expected -2	Pass
TC6.13	int2 specific test cases	int2 (4)	-23'sd7	hex 0x3F000000	8'd0	0xFE (-2)	1	-3.5 tie -> -4, expected -2, overflow	Pass
TC7.1	uint2 specific test cases	uint2 (5)	23'sd0	hex 0x3F800000	8'd0	0x00 (0)	0	exact low edge, expected 0	Pass
TC7.2	uint2 specific test cases	uint2 (5)	23'sd3	hex 0x3F800000	8'd0	0x03 (3)	0	exact high edge, expected 3	Pass
TC7.3	uint2 specific test cases	uint2 (5)	23'sd4	hex 0x3F800000	8'd0	0x03 (3)	1	one above high edge, expected 3, overflow	Pass
TC7.4	uint2 specific test cases	uint2 (5)	-23'sd1	hex 0x3F800000	8'd0	0x00 (0)	1	below low edge, expected 0, overflow	Pass
TC7.5	uint2 specific test cases	uint2 (5)	23'sd2	hex 0x3F800000	8'd1	0x03 (3)	0	2 + 1 = 3, expected 3	Pass
TC7.6	uint2 specific test cases	uint2 (5)	23'sd3	hex 0x3F800000	8'd1	0x03 (3)	1	3 + 1 = 4, expected 3, overflow	Pass
TC7.7	uint2 specific test cases	uint2 (5)	-23'sd1	hex 0x3F800000	8'd1	0x00 (0)	0	-1 + 1 = 0, expected 0	Pass
TC7.8	uint2 specific test cases	uint2 (5)	-23'sd2	hex 0x3F800000	8'd1	0x00 (0)	1	-2 + 1 = -1, expected 0, overflow	Pass
TC7.9	uint2 specific test cases	uint2 (5)	23'sd1	hex 0x3F000000	8'd0	0x00 (0)	0	0.5 tie, expected 0	Pass
TC7.10	uint2 specific test cases	uint2 (5)	23'sd3	hex 0x3F000000	8'd0	0x02 (2)	0	1.5 tie, expected 2	Pass
TC7.11	uint2 specific test cases	uint2 (5)	23'sd7	hex 0x3F000000	8'd0	0x03 (3)	1	3.5 tie -> 4, expected 3, overflow	Pass
TC7.12	uint2 specific test cases	uint2 (5)	-23'sd1	hex 0x3F000000	8'd0	0x00 (0)	0	-0.5 tie, expected 0	Pass
TC7.13	uint2 specific test cases	uint2 (5)	-23'sd3	hex 0x3F000000	8'd0	0x00 (0)	1	-1.5 tie, expected 0, overflow	Pass
TC8.1	Rounding boundaries	int8 (0)	23'sd8	hex 0x3D7FFFFF	8'sd0	0x00 (0)	0	int8 round P0a: < +0.5 -> 0	Pass
TC8.2	Rounding boundaries	int8 (0)	23'sd8	hex 0x3D800000	8'sd0	0x00 (0)	0	int8 round P0b: = +0.5 -> 0, tie to even	Pass
TC8.3	Rounding boundaries	int8 (0)	23'sd8	hex 0x3D800001	8'sd0	0x01 (1)	0	int8 round P0c: > +0.5 -> 1	Pass
TC8.4	Rounding boundaries	int8 (0)	23'sd4	hex 0x3EBFFFFF	8'sd0	0x01 (1)	0	int8 round P1a: < +1.5 -> 1	Pass
TC8.5	Rounding boundaries	int8 (0)	23'sd4	hex 0x3EC00000	8'sd0	0x02 (2)	0	int8 round P1b: = +1.5 -> 2, tie to even	Pass

TC8.6	Rounding boundaries	int8 (0)	23'sd4	hex 0x3EC00001	8'sd0	0x02 (2)	0	int8 round P1c: > +1.5 -> 2	Pass
TC8.7	Rounding boundaries	int8 (0)	-23'sd8	hex 0x3D7FFFFF	8'sd0	0x00 (0)	0	int8 round N0a: > -0.5 -> 0	Pass
TC8.8	Rounding boundaries	int8 (0)	-23'sd8	hex 0x3D800000	8'sd0	0x00 (0)	0	int8 round N0b: = -0.5 -> 0, tie to even	Pass
TC8.9	Rounding boundaries	int8 (0)	-23'sd8	hex 0x3D800001	8'sd0	0xFF (-1)	0	int8 round N0c: < -0.5 -> -1 = 8'hFF	Pass
TC8.10	Rounding boundaries	int8 (0)	-23'sd4	hex 0x3EBFFFFF	8'sd0	0xFF (-1)	0	int8 round N1a: > -1.5 -> -1 = 8'hFF	Pass
TC8.11	Rounding boundaries	int8 (0)	-23'sd4	hex 0x3EC00000	8'sd0	0xFE (-2)	0	int8 round N1b: = -1.5 -> -2 = 8'hFE, tie to even	Pass
TC8.12	Rounding boundaries	int8 (0)	-23'sd4	hex 0x3EC00001	8'sd0	0xFE (-2)	0	int8 round N1c: < -1.5 -> -2 = 8'hFE	Pass
TC8.13	Rounding boundaries	uint8 (1)	23'sd8	hex 0x3D7FFFFF	8'd0	0x00 (0)	0	uint8 round P0a: < +0.5 -> 0	Pass
TC8.14	Rounding boundaries	uint8 (1)	23'sd8	hex 0x3D800000	8'd0	0x00 (0)	0	uint8 round P0b: = +0.5 -> 0, tie to even	Pass
TC8.15	Rounding boundaries	uint8 (1)	23'sd8	hex 0x3D800001	8'd0	0x01 (1)	0	uint8 round P0c: > +0.5 -> 1	Pass
TC8.16	Rounding boundaries	uint8 (1)	23'sd4	hex 0x3EBFFFFF	8'd0	0x01 (1)	0	uint8 round P1a: < +1.5 -> 1	Pass
TC8.17	Rounding boundaries	uint8 (1)	23'sd4	hex 0x3EC00000	8'd0	0x02 (2)	0	uint8 round P1b: = +1.5 -> 2, tie to even	Pass
TC8.18	Rounding boundaries	uint8 (1)	23'sd4	hex 0x3EC00001	8'd0	0x02 (2)	0	uint8 round P1c: > +1.5 -> 2	Pass
TC8.19	Rounding boundaries	uint8 (1)	-23'sd8	hex 0x3D7FFFFF	8'd0	0x00 (0)	0	uint8 round N0a: > -0.5 -> 0	Pass
TC8.20	Rounding boundaries	uint8 (1)	-23'sd8	hex 0x3D800000	8'd0	0x00 (0)	0	uint8 round N0b: = -0.5 -> 0, tie to even	Pass
TC8.21	Rounding boundaries	uint8 (1)	-23'sd8	hex 0x3D800001	8'd0	0x00 (0)	1	uint8 round N0c: < -0.5 -> -1, clip to 0, overflow	Pass
TC8.22	Rounding boundaries	uint8 (1)	-23'sd4	hex 0x3EBFFFFF	8'd0	0x00 (0)	1	uint8 round N1a: > -1.5 -> -1, clip to 0, overflow	Pass
TC8.23	Rounding boundaries	uint8 (1)	-23'sd4	hex 0x3EC00000	8'd0	0x00 (0)	1	uint8 round N1b: = -1.5 -> -2, clip to 0, overflow	Pass
TC8.24	Rounding boundaries	uint8 (1)	-23'sd4	hex 0x3EC00001	8'd0	0x00 (0)	1	uint8 round N1c: < -1.5 -> -2, clip to 0, overflow	Pass
TC8.25	Rounding boundaries	int4 (2)	23'sd8	hex 0x3D7FFFFF	8'sd0	0x00 (0)	0	int4 round P0a: < +0.5 -> 0	Pass
TC8.26	Rounding boundaries	int4 (2)	23'sd8	hex 0x3D800000	8'sd0	0x00 (0)	0	int4 round P0b: = +0.5 -> 0, tie to even	Pass
TC8.27	Rounding boundaries	int4 (2)	23'sd8	hex 0x3D800001	8'sd0	0x01 (1)	0	int4 round P0c: > +0.5 -> 1	Pass
TC8.28	Rounding boundaries	int4 (2)	23'sd4	hex 0x3EBFFFFF	8'sd0	0x01 (1)	0	int4 round P1a: < +1.5 -> 1	Pass
TC8.29	Rounding boundaries	int4 (2)	23'sd4	hex 0x3EC00000	8'sd0	0x02 (2)	0	int4 round P1b: = +1.5 -> 2, tie to even	Pass
TC8.30	Rounding boundaries	int4 (2)	23'sd4	hex 0x3EC00001	8'sd0	0x02 (2)	0	int4 round P1c: > +1.5 -> 2	Pass
TC8.31	Rounding boundaries	int4 (2)	-23'sd8	hex 0x3D7FFFFF	8'sd0	0x00 (0)	0	int4 round N0a: > -0.5 -> 0	Pass
TC8.32	Rounding boundaries	int4 (2)	-23'sd8	hex 0x3D800000	8'sd0	0x00 (0)	0	int4 round N0b: = -0.5 -> 0, tie to even	Pass
TC8.33	Rounding boundaries	int4 (2)	-23'sd8	hex 0x3D800001	8'sd0	0xFF (-1)	0	int4 round N0c: < -0.5 -> -1 = 8'hFF	Pass
TC8.34	Rounding boundaries	int4 (2)	-23'sd4	hex 0x3EBFFFFF	8'sd0	0xFF (-1)	0	int4 round N1a: > -1.5 -> -1 = 8'hFF	Pass
TC8.35	Rounding boundaries	int4 (2)	-23'sd4	hex 0x3EC00000	8'sd0	0xFE (-2)	0	int4 round N1b: = -1.5 -> -2 = 8'hFE, tie to even	Pass
TC8.36	Rounding boundaries	int4 (2)	-23'sd4	hex 0x3EC00001	8'sd0	0xFE (-2)	0	int4 round N1c: < -1.5 -> -2 = 8'hFE	Pass
TC8.37	Rounding boundaries	uint4 (3)	23'sd8	hex 0x3D7FFFFF	8'd0	0x00 (0)	0	uint4 round P0a: < +0.5 -> 0	Pass
TC8.38	Rounding boundaries	uint4 (3)	23'sd8	hex 0x3D800000	8'd0	0x00 (0)	0	uint4 round P0b: = +0.5 -> 0, tie to even	Pass
TC8.39	Rounding boundaries	uint4 (3)	23'sd8	hex 0x3D800001	8'd0	0x01 (1)	0	uint4 round P0c: > +0.5 -> 1	Pass
TC8.40	Rounding boundaries	uint4 (3)	23'sd4	hex 0x3EBFFFFF	8'd0	0x01 (1)	0	uint4 round P1a: < +1.5 -> 1	Pass
TC8.41	Rounding boundaries	uint4 (3)	23'sd4	hex 0x3EC00000	8'd0	0x02 (2)	0	uint4 round P1b: = +1.5 -> 2, tie to even	Pass
TC8.42	Rounding boundaries	uint4 (3)	23'sd4	hex 0x3EC00001	8'd0	0x02 (2)	0	uint4 round P1c: > +1.5 -> 2	Pass
TC8.43	Rounding boundaries	uint4 (3)	-23'sd8	hex 0x3D7FFFFF	8'd0	0x00 (0)	0	uint4 round N0a: > -0.5 -> 0	Pass
TC8.44	Rounding boundaries	uint4 (3)	-23'sd8	hex 0x3D800000	8'd0	0x00 (0)	0	uint4 round N0b: = -0.5 -> 0, tie to even	Pass
TC8.45	Rounding boundaries	uint4 (3)	-23'sd8	hex 0x3D800001	8'd0	0x00 (0)	1	uint4 round N0c: < -0.5 -> -1, clip to 0, overflow	Pass
TC8.46	Rounding boundaries	uint4 (3)	-23'sd4	hex 0x3EBFFFFF	8'd0	0x00 (0)	1	uint4 round N1a: > -1.5 -> -1, clip to 0, overflow	Pass
TC8.47	Rounding boundaries	uint4 (3)	-23'sd4	hex 0x3EC00000	8'd0	0x00 (0)	1	uint4 round N1b: = -1.5 -> -2, clip to 0, overflow	Pass
TC8.48	Rounding boundaries	uint4 (3)	-23'sd4	hex 0x3EC00001	8'd0	0x00 (0)	1	uint4 round N1c: < -1.5 -> -2, clip to 0, overflow	Pass

TC8.49	Rounding boundaries	int2 (4)	23'sd8	hex 0x3D7FFFFF	8'sd0	0x00 (0)	0	int2 round P0a: < +0.5 -> 0	Pass
TC8.50	Rounding boundaries	int2 (4)	23'sd8	hex 0x3D800000	8'sd0	0x00 (0)	0	int2 round P0b: = +0.5 -> 0, tie to even	Pass
TC8.51	Rounding boundaries	int2 (4)	23'sd8	hex 0x3D800001	8'sd0	0x01 (1)	0	int2 round P0c: > +0.5 -> 1	Pass
TC8.52	Rounding boundaries	int2 (4)	23'sd4	hex 0x3EBFFFFF	8'sd0	0x01 (1)	0	int2 round P1a: < +1.5 -> 1	Pass
TC8.53	Rounding boundaries	int2 (4)	23'sd4	hex 0x3EC00000	8'sd0	0x01 (1)	1	int2 round P1b: = +1.5 -> 2, clip to 1, overflow	Pass
TC8.54	Rounding boundaries	int2 (4)	23'sd4	hex 0x3EC00001	8'sd0	0x01 (1)	1	int2 round P1c: > +1.5 -> 2, clip to 1, overflow	Pass
TC8.55	Rounding boundaries	int2 (4)	-23'sd8	hex 0x3D7FFFFF	8'sd0	0x00 (0)	0	int2 round N0a: > -0.5 -> 0	Pass
TC8.56	Rounding boundaries	int2 (4)	-23'sd8	hex 0x3D800000	8'sd0	0x00 (0)	0	int2 round N0b: = -0.5 -> 0, tie to even	Pass
TC8.57	Rounding boundaries	int2 (4)	-23'sd8	hex 0x3D800001	8'sd0	0xFF (-1)	0	int2 round N0c: < -0.5 -> -1 = 8'hFF	Pass
TC8.58	Rounding boundaries	int2 (4)	-23'sd4	hex 0x3EBFFFFF	8'sd0	0xFF (-1)	0	int2 round N1a: > -1.5 -> -1 = 8'hFF	Pass
TC8.59	Rounding boundaries	int2 (4)	-23'sd4	hex 0x3EC00000	8'sd0	0xFE (-2)	0	int2 round N1b: = -1.5 -> -2 = 8'hFE, tie to even	Pass
TC8.60	Rounding boundaries	int2 (4)	-23'sd4	hex 0x3EC00001	8'sd0	0xFE (-2)	0	int2 round N1c: < -1.5 -> -2 = 8'hFE	Pass
TC8.61	Rounding boundaries	uint2 (5)	23'sd8	hex 0x3D7FFFFF	8'd0	0x00 (0)	0	uint2 round P0a: < +0.5 -> 0	Pass
TC8.62	Rounding boundaries	uint2 (5)	23'sd8	hex 0x3D800000	8'd0	0x00 (0)	0	uint2 round P0b: = +0.5 -> 0, tie to even	Pass
TC8.63	Rounding boundaries	uint2 (5)	23'sd8	hex 0x3D800001	8'd0	0x01 (1)	0	uint2 round P0c: > +0.5 -> 1	Pass
TC8.64	Rounding boundaries	uint2 (5)	23'sd4	hex 0x3EBFFFFF	8'd0	0x01 (1)	0	uint2 round P1a: < +1.5 -> 1	Pass
TC8.65	Rounding boundaries	uint2 (5)	23'sd4	hex 0x3EC00000	8'd0	0x02 (2)	0	uint2 round P1b: = +1.5 -> 2, tie to even	Pass
TC8.66	Rounding boundaries	uint2 (5)	23'sd4	hex 0x3EC00001	8'd0	0x02 (2)	0	uint2 round P1c: > +1.5 -> 2	Pass
TC8.67	Rounding boundaries	uint2 (5)	-23'sd8	hex 0x3D7FFFFF	8'd0	0x00 (0)	0	uint2 round N0a: > -0.5 -> 0	Pass
TC8.68	Rounding boundaries	uint2 (5)	-23'sd8	hex 0x3D800000	8'd0	0x00 (0)	0	uint2 round N0b: = -0.5 -> 0, tie to even	Pass
TC8.69	Rounding boundaries	uint2 (5)	-23'sd8	hex 0x3D800001	8'd0	0x00 (0)	1	uint2 round N0c: < -0.5 -> -1, clip to 0, overflow	Pass
TC8.70	Rounding boundaries	uint2 (5)	-23'sd4	hex 0x3EBFFFFF	8'd0	0x00 (0)	1	uint2 round N1a: > -1.5 -> -1, clip to 0, overflow	Pass
TC8.71	Rounding boundaries	uint2 (5)	-23'sd4	hex 0x3EC00000	8'd0	0x00 (0)	1	uint2 round N1b: = -1.5 -> -2, clip to 0, overflow	Pass
TC8.72	Rounding boundaries	uint2 (5)	-23'sd4	hex 0x3EC00001	8'd0	0x00 (0)	1	uint2 round N1c: < -1.5 -> -2, clip to 0, overflow	Pass
TC9.1	Shift	uint8 (1)	23'sd4194303	hex 0x32FFFFFF	8'd44	0x2C (44)	0	shifts=101, below supported shift range, expected 44	Pass
TC9.2	Shift	uint8 (1)	23'sd4194303	hex 0x337FFFFF	8'd44	0x2C (44)	0	shifts=102, smallest supported shift, product still rounds to 0, expected 44	Pass
TC9.3	Shift	uint8 (1)	23'sd4194303	hex 0x347FFFFF	8'd44	0x2D (45)	0	shifts=104 with large mantissa, product rounds to 1, expected 45	Pass
TC9.4	Shift	uint8 (1)	23'sd4194303	hex 0x34800000	8'd44	0x2D (45)	0	shifts=105, scale=2^-22, product rounds to 1, expected 45	Pass
TC9.5	Shift	uint8 (1)	23'sd1	hex 0x3F000000	8'd44	0x2C (44)	0	shifts=126, scale=0.5, tie rounds to 0, expected 44	Pass
TC9.6	Shift	uint8 (1)	23'sd1	hex 0x3F800000	8'd44	0x2D (45)	0	shifts=127, scale=1.0, expected 45	Pass
TC9.7	Shift	uint8 (1)	23'sd64	hex 0x40000000	8'd44	0xAC (172)	0	shifts=128, scale=2.0, 64*2+44=172, expected 172	Pass
TC9.8	Shift	uint8 (1)	23'sd1	hex 0x43000000	8'd0	0x80 (128)	0	shifts=134, largest normal shift, 1*128=128, expected 128, no overflow for uint8	Pass
TC9.9	Shift	uint8 (1)	23'sd1	hex 0x43800000	8'd44	0xFF (255)	1	shifts=135, above normal range, expected 255, overflow; zero point is bypassed on overflow_1	Pass
TC9.10	Shift	uint8 (1)	-23'sd1	hex 0x43800000	8'd44	0x00 (0)	1	shifts=135 negative, expected 0, overflow for uint8	Pass
TC9.11	Shift	int8 (0)	23'sd1	hex 0x43800000	8'd44	0x7F (127)	1	shifts=135 positive, expected 127, overflow for int8	Pass
TC9.12	Shift	int8 (0)	-23'sd1	hex 0x43800000	8'd44	0x80 (-128)	1	shifts=135 negative, expected -128 = 8'h80, overflow for int8	Pass
TC10.1	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x37FFFFFF	8'd44	0x7F (127)	1	sweep P1: 128 + 44 = 172 -> 127, overflow	Pass
TC10.2	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x377FFFFF	8'd44	0x6C (108)	0	sweep P2: 64 + 44 = 108	Pass
TC10.3	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x36FFFFFF	8'd44	0x4C (76)	0	sweep P3: 32 + 44 = 76	Pass
TC10.4	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x367FFFFF	8'd44	0x3C (60)	0	sweep P4: 16 + 44 = 60	Pass

TC10.5	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x35FFFFFF	8'd44	0x34 (52)	0	sweep P5: 8 + 44 = 52	Pass
TC10.6	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x357FFFFF	8'd44	0x30 (48)	0	sweep P6: 4 + 44 = 48	Pass
TC10.7	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x34FFFFFF	8'd44	0x2E (46)	0	sweep P7: 2 + 44 = 46	Pass
TC10.8	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x347FFFFF	8'd44	0x2D (45)	0	sweep P8: 1 + 44 = 45	Pass
TC10.9	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x33FFFFFF	8'd44	0x2C (44)	0	sweep P9: 0 + 44 = 44	Pass
TC10.10	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x337FFFFF	8'd44	0x2C (44)	0	sweep P10: 0 + 44 = 44	Pass
TC10.11	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x32FFFFFF	8'd44	0x2C (44)	0	sweep P11: 0 + 44 = 44	Pass
TC10.12	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x327FFFFF	8'd44	0x2C (44)	0	sweep P12: 0 + 44 = 44	Pass
TC10.13	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x31FFFFFF	8'd44	0x2C (44)	0	sweep P13: 0 + 44 = 44	Pass
TC10.14	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x317FFFFF	8'd44	0x2C (44)	0	sweep P14: 0 + 44 = 44	Pass
TC10.15	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x30FFFFFF	8'd44	0x2C (44)	0	sweep P15: 0 + 44 = 44	Pass
TC10.16	int8 scale sweep	int8 (0)	23'sd4194303	hex 0x307FFFFF	8'd44	0x2C (44)	0	sweep P16: 0 + 44 = 44	Pass
TC10.17	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x37FFFFFF	8'd44	0xAC (-84)	0	sweep N1: -128 + 44 = -84 = 8'hAC	Pass
TC10.18	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x377FFFFF	8'd44	0xEC (-20)	0	sweep N2: -64 + 44 = -20 = 8'hEC	Pass
TC10.19	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x36FFFFFF	8'd44	0x0C (12)	0	sweep N3: -32 + 44 = 12	Pass
TC10.20	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x367FFFFF	8'd44	0x1C (28)	0	sweep N4: -16 + 44 = 28	Pass
TC10.21	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x35FFFFFF	8'd44	0x24 (36)	0	sweep N5: -8 + 44 = 36	Pass
TC10.22	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x357FFFFF	8'd44	0x28 (40)	0	sweep N6: -4 + 44 = 40	Pass
TC10.23	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x34FFFFFF	8'd44	0x2A (42)	0	sweep N7: -2 + 44 = 42	Pass
TC10.24	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x347FFFFF	8'd44	0x2B (43)	0	sweep N8: -1 + 44 = 43	Pass
TC10.25	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x33FFFFFF	8'd44	0x2C (44)	0	sweep N9: 0 + 44 = 44	Pass
TC10.26	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x337FFFFF	8'd44	0x2C (44)	0	sweep N10: 0 + 44 = 44	Pass
TC10.27	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x32FFFFFF	8'd44	0x2C (44)	0	sweep N11: 0 + 44 = 44	Pass
TC10.28	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x327FFFFF	8'd44	0x2C (44)	0	sweep N12: 0 + 44 = 44	Pass
TC10.29	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x31FFFFFF	8'd44	0x2C (44)	0	sweep N13: 0 + 44 = 44	Pass
TC10.30	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x317FFFFF	8'd44	0x2C (44)	0	sweep N14: 0 + 44 = 44	Pass
TC10.31	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x30FFFFFF	8'd44	0x2C (44)	0	sweep N15: 0 + 44 = 44	Pass
TC10.32	int8 scale sweep	int8 (0)	-23'sd4194303	hex 0x307FFFFF	8'd44	0x2C (44)	0	sweep N16: 0 + 44 = 44	Pass
TC11.1	uint8 scale sweep	uint8 (1)	23'sd4194303	hex 0x37FFFFFF	8'd44	0xAC (172)	0	uint8 sweep P1: 128 + 44 = 172	Pass
TC11.2	uint8 scale sweep	uint8 (1)	23'sd4194303	hex 0x377FFFFF	8'd44	0x6C (108)	0	uint8 sweep P2: 64 + 44 = 108	Pass
TC11.3	uint8 scale sweep	uint8 (1)	23'sd4194303	hex 0x36FFFFFF	8'd44	0x4C (76)	0	uint8 sweep P3: 32 + 44 = 76	Pass
TC11.4	uint8 scale sweep	uint8 (1)	23'sd4194303	hex 0x367FFFFF	8'd44	0x3C (60)	0	uint8 sweep P4: 16 + 44 = 60	Pass
TC11.5	uint8 scale sweep	uint8 (1)	23'sd4194303	hex 0x35FFFFFF	8'd44	0x34 (52)	0	uint8 sweep P5: 8 + 44 = 52	Pass
TC11.6	uint8 scale sweep	uint8 (1)	23'sd4194303	hex 0x357FFFFF	8'd44	0x30 (48)	0	uint8 sweep P6: 4 + 44 = 48	Pass
TC11.7	uint8 scale sweep	uint8 (1)	23'sd4194303	hex 0x34FFFFFF	8'd44	0x2E (46)	0	uint8 sweep P7: 2 + 44 = 46	Pass
TC11.8	uint8 scale sweep	uint8 (1)	23'sd4194303	hex 0x347FFFFF	8'd44	0x2D (45)	0	uint8 sweep P8: 1 + 44 = 45	Pass
TC11.9	uint8 scale sweep	uint8 (1)	23'sd4194303	hex 0x33FFFFFF	8'd44	0x2C (44)	0	uint8 sweep P9: 0 + 44 = 44	Pass
TC11.10	uint8 scale sweep	uint8 (1)	23'sd4194303	hex 0x337FFFFF	8'd44	0x2C (44)	0	uint8 sweep P10: 0 + 44 = 44	Pass
TC11.11	uint8 scale sweep	uint8 (1)	-23'sd4194303	hex 0x37FFFFFF	8'd44	0x00 (0)	1	uint8 sweep N1: -128 + 44 = -84 -> 0, overflow	Pass
TC11.12	uint8 scale sweep	uint8 (1)	-23'sd4194303	hex 0x377FFFFF	8'd44	0x00 (0)	1	uint8 sweep N2: -64 + 44 = -20 -> 0, overflow	Pass
TC11.13	uint8 scale sweep	uint8 (1)	-23'sd4194303	hex 0x36FFFFFF	8'd44	0x0C (12)	0	uint8 sweep N3: -32 + 44 = 12	Pass
TC11.14	uint8 scale sweep	uint8 (1)	-23'sd4194303	hex 0x367FFFFF	8'd44	0x1C (28)	0	uint8 sweep N4: -16 + 44 = 28	Pass
TC11.15	uint8 scale sweep	uint8 (1)	-23'sd4194303	hex 0x35FFFFFF	8'd44	0x24 (36)	0	uint8 sweep N5: -8 + 44 = 36	Pass

TC11.16	uint8 scale sweep	uint8 (1)	-23'sd4194303	hex 0x357FFFFF	8'd44	0x28 (40)	0	uint8 sweep N6: -4 + 44 = 40	Pass
TC11.17	uint8 scale sweep	uint8 (1)	-23'sd4194303	hex 0x34FFFFFF	8'd44	0x2A (42)	0	uint8 sweep N7: -2 + 44 = 42	Pass
TC11.18	uint8 scale sweep	uint8 (1)	-23'sd4194303	hex 0x347FFFFF	8'd44	0x2B (43)	0	uint8 sweep N8: -1 + 44 = 43	Pass
TC11.19	uint8 scale sweep	uint8 (1)	-23'sd4194303	hex 0x33FFFFFF	8'd44	0x2C (44)	0	uint8 sweep N9: 0 + 44 = 44	Pass
TC11.20	uint8 scale sweep	uint8 (1)	-23'sd4194303	hex 0x337FFFFF	8'd44	0x2C (44)	0	uint8 sweep N10: 0 + 44 = 44	Pass
TC12.1	int4 scale sweep	int4 (2)	23'sd4194303	hex 0x37FFFFFF	8'd4	0x07 (7)	1	int4 sweep P1: 128 + 4 = 132 -> 7, overflow	Pass
TC12.2	int4 scale sweep	int4 (2)	23'sd4194303	hex 0x377FFFFF	8'd4	0x07 (7)	1	int4 sweep P2: 64 + 4 = 68 -> 7, overflow	Pass
TC12.3	int4 scale sweep	int4 (2)	23'sd4194303	hex 0x36FFFFFF	8'd4	0x07 (7)	1	int4 sweep P3: 32 + 4 = 36 -> 7, overflow	Pass
TC12.4	int4 scale sweep	int4 (2)	23'sd4194303	hex 0x367FFFFF	8'd4	0x07 (7)	1	int4 sweep P4: 16 + 4 = 20 -> 7, overflow	Pass
TC12.5	int4 scale sweep	int4 (2)	23'sd4194303	hex 0x35FFFFFF	8'd4	0x07 (7)	1	int4 sweep P5: 8 + 4 = 12 -> 7, overflow	Pass
TC12.6	int4 scale sweep	int4 (2)	23'sd4194303	hex 0x357FFFFF	8'd4	0x07 (7)	1	int4 sweep P6: 4 + 4 = 8 -> 7, overflow	Pass
TC12.7	int4 scale sweep	int4 (2)	23'sd4194303	hex 0x34FFFFFF	8'd4	0x06 (6)	0	int4 sweep P7: 2 + 4 = 6	Pass
TC12.8	int4 scale sweep	int4 (2)	23'sd4194303	hex 0x347FFFFF	8'd4	0x05 (5)	0	int4 sweep P8: 1 + 4 = 5	Pass
TC12.9	int4 scale sweep	int4 (2)	23'sd4194303	hex 0x33FFFFFF	8'd4	0x04 (4)	0	int4 sweep P9: 0 + 4 = 4	Pass
TC12.10	int4 scale sweep	int4 (2)	23'sd4194303	hex 0x337FFFFF	8'd4	0x04 (4)	0	int4 sweep P10: 0 + 4 = 4	Pass
TC12.11	int4 scale sweep	int4 (2)	-23'sd4194303	hex 0x37FFFFFF	8'd4	0xF8 (-8)	1	int4 sweep N1: -128 + 4 = -124 -> -8 = 8'hF8, overflow	Pass
TC12.12	int4 scale sweep	int4 (2)	-23'sd4194303	hex 0x377FFFFF	8'd4	0xF8 (-8)	1	int4 sweep N2: -64 + 4 = -60 -> -8 = 8'hF8, overflow	Pass
TC12.13	int4 scale sweep	int4 (2)	-23'sd4194303	hex 0x36FFFFFF	8'd4	0xF8 (-8)	1	int4 sweep N3: -32 + 4 = -28 -> -8 = 8'hF8, overflow	Pass
TC12.14	int4 scale sweep	int4 (2)	-23'sd4194303	hex 0x367FFFFF	8'd4	0xF8 (-8)	1	int4 sweep N4: -16 + 4 = -12 -> -8 = 8'hF8, overflow	Pass
TC12.15	int4 scale sweep	int4 (2)	-23'sd4194303	hex 0x35FFFFFF	8'd4	0xFC (-4)	0	int4 sweep N5: -8 + 4 = -4 = 8'hFC	Pass
TC12.16	int4 scale sweep	int4 (2)	-23'sd4194303	hex 0x357FFFFF	8'd4	0x00 (0)	0	int4 sweep N6: -4 + 4 = 0	Pass
TC12.17	int4 scale sweep	int4 (2)	-23'sd4194303	hex 0x34FFFFFF	8'd4	0x02 (2)	0	int4 sweep N7: -2 + 4 = 2	Pass
TC12.18	int4 scale sweep	int4 (2)	-23'sd4194303	hex 0x347FFFFF	8'd4	0x03 (3)	0	int4 sweep N8: -1 + 4 = 3	Pass
TC12.19	int4 scale sweep	int4 (2)	-23'sd4194303	hex 0x33FFFFFF	8'd4	0x04 (4)	0	int4 sweep N9: 0 + 4 = 4	Pass
TC12.20	int4 scale sweep	int4 (2)	-23'sd4194303	hex 0x337FFFFF	8'd4	0x04 (4)	0	int4 sweep N10: 0 + 4 = 4	Pass
TC13.1	uint4 scale sweep	uint4 (3)	23'sd4194303	hex 0x37FFFFFF	8'd4	0x0F (15)	1	uint4 sweep P1: 128 + 4 = 132 -> 15, overflow	Pass
TC13.2	uint4 scale sweep	uint4 (3)	23'sd4194303	hex 0x377FFFFF	8'd4	0x0F (15)	1	uint4 sweep P2: 64 + 4 = 68 -> 15, overflow	Pass
TC13.3	uint4 scale sweep	uint4 (3)	23'sd4194303	hex 0x36FFFFFF	8'd4	0x0F (15)	1	uint4 sweep P3: 32 + 4 = 36 -> 15, overflow	Pass
TC13.4	uint4 scale sweep	uint4 (3)	23'sd4194303	hex 0x367FFFFF	8'd4	0x0F (15)	1	uint4 sweep P4: 16 + 4 = 20 -> 15, overflow	Pass
TC13.5	uint4 scale sweep	uint4 (3)	23'sd4194303	hex 0x35FFFFFF	8'd4	0x0C (12)	0	uint4 sweep P5: 8 + 4 = 12	Pass
TC13.6	uint4 scale sweep	uint4 (3)	23'sd4194303	hex 0x357FFFFF	8'd4	0x08 (8)	0	uint4 sweep P6: 4 + 4 = 8	Pass
TC13.7	uint4 scale sweep	uint4 (3)	23'sd4194303	hex 0x34FFFFFF	8'd4	0x06 (6)	0	uint4 sweep P7: 2 + 4 = 6	Pass
TC13.8	uint4 scale sweep	uint4 (3)	23'sd4194303	hex 0x347FFFFF	8'd4	0x05 (5)	0	uint4 sweep P8: 1 + 4 = 5	Pass
TC13.9	uint4 scale sweep	uint4 (3)	23'sd4194303	hex 0x33FFFFFF	8'd4	0x04 (4)	0	uint4 sweep P9: 0 + 4 = 4	Pass
TC13.10	uint4 scale sweep	uint4 (3)	23'sd4194303	hex 0x337FFFFF	8'd4	0x04 (4)	0	uint4 sweep P10: 0 + 4 = 4	Pass
TC13.11	uint4 scale sweep	uint4 (3)	-23'sd4194303	hex 0x37FFFFFF	8'd4	0x00 (0)	1	uint4 sweep N1: -128 + 4 = -124 -> 0, overflow	Pass
TC13.12	uint4 scale sweep	uint4 (3)	-23'sd4194303	hex 0x377FFFFF	8'd4	0x00 (0)	1	uint4 sweep N2: -64 + 4 = -60 -> 0, overflow	Pass
TC13.13	uint4 scale sweep	uint4 (3)	-23'sd4194303	hex 0x36FFFFFF	8'd4	0x00 (0)	1	uint4 sweep N3: -32 + 4 = -28 -> 0, overflow	Pass
TC13.14	uint4 scale sweep	uint4 (3)	-23'sd4194303	hex 0x367FFFFF	8'd4	0x00 (0)	1	uint4 sweep N4: -16 + 4 = -12 -> 0, overflow	Pass
TC13.15	uint4 scale sweep	uint4 (3)	-23'sd4194303	hex 0x35FFFFFF	8'd4	0x00 (0)	1	uint4 sweep N5: -8 + 4 = -4 -> 0, overflow	Pass
TC13.16	uint4 scale sweep	uint4 (3)	-23'sd4194303	hex 0x357FFFFF	8'd4	0x00 (0)	0	uint4 sweep N6: -4 + 4 = 0	Pass
TC13.17	uint4 scale sweep	uint4 (3)	-23'sd4194303	hex 0x34FFFFFF	8'd4	0x02 (2)	0	uint4 sweep N7: -2 + 4 = 2	Pass
TC13.18	uint4 scale sweep	uint4 (3)	-23'sd4194303	hex 0x347FFFFF	8'd4	0x03 (3)	0	uint4 sweep N8: -1 + 4 = 3	Pass

TC13.19	uint4 scale sweep	uint4 (3)	-23'sd4194303	hex 0x33FFFFFF	8'd4	0x04 (4)	0	uint4 sweep N9: 0 + 4 = 4	Pass
TC13.20	uint4 scale sweep	uint4 (3)	-23'sd4194303	hex 0x337FFFFF	8'd4	0x04 (4)	0	uint4 sweep N10: 0 + 4 = 4	Pass
TC14.1	int2 scale sweep	int2 (4)	23'sd4194303	hex 0x37FFFFFF	8'd0	0x01 (1)	1	int2 sweep P1: 128 -> 1, overflow	Pass
TC14.2	int2 scale sweep	int2 (4)	23'sd4194303	hex 0x377FFFFF	8'd0	0x01 (1)	1	int2 sweep P2: 64 -> 1, overflow	Pass
TC14.3	int2 scale sweep	int2 (4)	23'sd4194303	hex 0x36FFFFFF	8'd0	0x01 (1)	1	int2 sweep P3: 32 -> 1, overflow	Pass
TC14.4	int2 scale sweep	int2 (4)	23'sd4194303	hex 0x367FFFFF	8'd0	0x01 (1)	1	int2 sweep P4: 16 -> 1, overflow	Pass
TC14.5	int2 scale sweep	int2 (4)	23'sd4194303	hex 0x35FFFFFF	8'd0	0x01 (1)	1	int2 sweep P5: 8 -> 1, overflow	Pass
TC14.6	int2 scale sweep	int2 (4)	23'sd4194303	hex 0x357FFFFF	8'd0	0x01 (1)	1	int2 sweep P6: 4 -> 1, overflow	Pass
TC14.7	int2 scale sweep	int2 (4)	23'sd4194303	hex 0x34FFFFFF	8'd0	0x01 (1)	1	int2 sweep P7: 2 -> 1, overflow	Pass
TC14.8	int2 scale sweep	int2 (4)	23'sd4194303	hex 0x347FFFFF	8'd0	0x01 (1)	0	int2 sweep P8: 1 -> 1	Pass
TC14.9	int2 scale sweep	int2 (4)	23'sd4194303	hex 0x33FFFFFF	8'd0	0x00 (0)	0	int2 sweep P9: 0 -> 0	Pass
TC14.10	int2 scale sweep	int2 (4)	23'sd4194303	hex 0x337FFFFF	8'd0	0x00 (0)	0	int2 sweep P10: 0 -> 0	Pass
TC14.11	int2 scale sweep	int2 (4)	-23'sd4194303	hex 0x37FFFFFF	8'd0	0xFE (-2)	1	int2 sweep N1: -128 -> -2 = 8'hFE, overflow	Pass
TC14.12	int2 scale sweep	int2 (4)	-23'sd4194303	hex 0x377FFFFF	8'd0	0xFE (-2)	1	int2 sweep N2: -64 -> -2 = 8'hFE, overflow	Pass
TC14.13	int2 scale sweep	int2 (4)	-23'sd4194303	hex 0x36FFFFFF	8'd0	0xFE (-2)	1	int2 sweep N3: -32 -> -2 = 8'hFE, overflow	Pass
TC14.14	int2 scale sweep	int2 (4)	-23'sd4194303	hex 0x367FFFFF	8'd0	0xFE (-2)	1	int2 sweep N4: -16 -> -2 = 8'hFE, overflow	Pass
TC14.15	int2 scale sweep	int2 (4)	-23'sd4194303	hex 0x35FFFFFF	8'd0	0xFE (-2)	1	int2 sweep N5: -8 -> -2 = 8'hFE, overflow	Pass
TC14.16	int2 scale sweep	int2 (4)	-23'sd4194303	hex 0x357FFFFF	8'd0	0xFE (-2)	1	int2 sweep N6: -4 -> -2 = 8'hFE, overflow	Pass
TC14.17	int2 scale sweep	int2 (4)	-23'sd4194303	hex 0x34FFFFFF	8'd0	0xFE (-2)	0	int2 sweep N7: -2 -> -2 = 8'hFE	Pass
TC14.18	int2 scale sweep	int2 (4)	-23'sd4194303	hex 0x347FFFFF	8'd0	0xFF (-1)	0	int2 sweep N8: -1 -> -1 = 8'hFF	Pass
TC14.19	int2 scale sweep	int2 (4)	-23'sd4194303	hex 0x33FFFFFF	8'd0	0x00 (0)	0	int2 sweep N9: 0 -> 0	Pass
TC14.20	int2 scale sweep	int2 (4)	-23'sd4194303	hex 0x337FFFFF	8'd0	0x00 (0)	0	int2 sweep N10: 0 -> 0	Pass
TC15.1	uint2 scale sweep	uint2 (5)	23'sd4194303	hex 0x37FFFFFF	8'd1	0x03 (3)	1	uint2 sweep P1: 128 + 1 = 129 -> 3, overflow	Pass
TC15.2	uint2 scale sweep	uint2 (5)	23'sd4194303	hex 0x377FFFFF	8'd1	0x03 (3)	1	uint2 sweep P2: 64 + 1 = 65 -> 3, overflow	Pass
TC15.3	uint2 scale sweep	uint2 (5)	23'sd4194303	hex 0x36FFFFFF	8'd1	0x03 (3)	1	uint2 sweep P3: 32 + 1 = 33 -> 3, overflow	Pass
TC15.4	uint2 scale sweep	uint2 (5)	23'sd4194303	hex 0x367FFFFF	8'd1	0x03 (3)	1	uint2 sweep P4: 16 + 1 = 17 -> 3, overflow	Pass
TC15.5	uint2 scale sweep	uint2 (5)	23'sd4194303	hex 0x35FFFFFF	8'd1	0x03 (3)	1	uint2 sweep P5: 8 + 1 = 9 -> 3, overflow	Pass
TC15.6	uint2 scale sweep	uint2 (5)	23'sd4194303	hex 0x357FFFFF	8'd1	0x03 (3)	1	uint2 sweep P6: 4 + 1 = 5 -> 3, overflow	Pass
TC15.7	uint2 scale sweep	uint2 (5)	23'sd4194303	hex 0x34FFFFFF	8'd1	0x03 (3)	0	uint2 sweep P7: 2 + 1 = 3	Pass
TC15.8	uint2 scale sweep	uint2 (5)	23'sd4194303	hex 0x347FFFFF	8'd1	0x02 (2)	0	uint2 sweep P8: 1 + 1 = 2	Pass
TC15.9	uint2 scale sweep	uint2 (5)	23'sd4194303	hex 0x33FFFFFF	8'd1	0x01 (1)	0	uint2 sweep P9: 0 + 1 = 1	Pass
TC15.10	uint2 scale sweep	uint2 (5)	23'sd4194303	hex 0x337FFFFF	8'd1	0x01 (1)	0	uint2 sweep P10: 0 + 1 = 1	Pass
TC15.11	uint2 scale sweep	uint2 (5)	-23'sd4194303	hex 0x37FFFFFF	8'd1	0x00 (0)	1	uint2 sweep N1: -128 + 1 = -127 -> 0, overflow	Pass
TC15.12	uint2 scale sweep	uint2 (5)	-23'sd4194303	hex 0x377FFFFF	8'd1	0x00 (0)	1	uint2 sweep N2: -64 + 1 = -63 -> 0, overflow	Pass
TC15.13	uint2 scale sweep	uint2 (5)	-23'sd4194303	hex 0x36FFFFFF	8'd1	0x00 (0)	1	uint2 sweep N3: -32 + 1 = -31 -> 0, overflow	Pass
TC15.14	uint2 scale sweep	uint2 (5)	-23'sd4194303	hex 0x367FFFFF	8'd1	0x00 (0)	1	uint2 sweep N4: -16 + 1 = -15 -> 0, overflow	Pass
TC15.15	uint2 scale sweep	uint2 (5)	-23'sd4194303	hex 0x35FFFFFF	8'd1	0x00 (0)	1	uint2 sweep N5: -8 + 1 = -7 -> 0, overflow	Pass
TC15.16	uint2 scale sweep	uint2 (5)	-23'sd4194303	hex 0x357FFFFF	8'd1	0x00 (0)	1	uint2 sweep N6: -4 + 1 = -3 -> 0, overflow	Pass
TC15.17	uint2 scale sweep	uint2 (5)	-23'sd4194303	hex 0x34FFFFFF	8'd1	0x00 (0)	1	uint2 sweep N7: -2 + 1 = -1 -> 0, overflow	Pass
TC15.18	uint2 scale sweep	uint2 (5)	-23'sd4194303	hex 0x347FFFFF	8'd1	0x00 (0)	0	uint2 sweep N8: -1 + 1 = 0	Pass
TC15.19	uint2 scale sweep	uint2 (5)	-23'sd4194303	hex 0x33FFFFFF	8'd1	0x01 (1)	0	uint2 sweep N9: 0 + 1 = 1	Pass
TC15.20	uint2 scale sweep	uint2 (5)	-23'sd4194303	hex 0x337FFFFF	8'd1	0x01 (1)	0	uint2 sweep N10: 0 + 1 = 1	Pass
TC16.1	DType equivalent	int8 (6)	23'sd127	hex 0x3F800000	8'd0	0x7F (127)	0	dtype 6 behaves like int8, expected 127	Pass

TC16.2	DType equivalent	int8 (6)	23'sd128	hex 0x3F800000	8'd0	0x7F (127)	1	dtype 6 positive saturation, expected 127, overflow	**Pass**
TC16.3	DType equivalent	int8 (6)	-23'sd128	hex 0x3F800000	8'd0	0x80 (-128)	0	dtype 6 negative edge, expected -128 = 8'h80	**Pass**
TC16.4	DType equivalent	int8 (6)	-23'sd129	hex 0x3F800000	8'd0	0x80 (-128)	1	dtype 6 negative saturation, expected -128, overflow	**Pass**
TC16.5	DType equivalent	uint8 (7)	23'sd255	hex 0x3F800000	8'd0	0xFF (255)	0	dtype 7 behaves like uint8, expected 255	**Pass**
TC16.6	DType equivalent	uint8 (7)	23'sd256	hex 0x3F800000	8'd0	0xFF (255)	1	dtype 7 positive saturation, expected 255, overflow	**Pass**
TC16.7	DType equivalent	uint8 (7)	-23'sd1	hex 0x3F800000	8'd0	0x00 (0)	1	dtype 7 negative saturation, expected 0, overflow	**Pass**